

what is slopware, how do you spot it, what do you do?

a curriculum that teaches you to measure the distance between something that looks like it works and something that actually holds up. every chapter ends with a check you run on your own project.

6 / 55 chapters · pdf · every published chapter on one page. for printing or the pdf.

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what should I call you?

this name stays in this browser. it is not sent anywhere.

why does this book exist?

Why this book exists, who is writing it, and how to read it.

I have been using forums such as Reddit, Quora and Ekşi Sözlük for years, and once I was one of the people posting there. People make fun of localhost:3000 users, and they believe CS is dead because somebody developed a local website. Actually, I hate needless jokes made just to fit in. But also, lack of technical depth gave those localhost:3000 owners the courage of a superhero. So instead of joining in, I decided to create a blog about how to avoid slopware, based on my own experiences.

Learning it cost me 53 repos, 8 pull requests and 6 Claude Max x20 accounts. It also cost me exploding token bills, sleepless nights, and the 5 kilos I lost to hunger and lack of sleep. I do not want that to happen to anybody else, and I do not want a power this large to stay hidden. I write this, and I am writing it with my own pain.

what should I call you?

I do not want this name to stay in this browser. it is not sent anywhere. I do not want there to be a C++, Python or Java language, I treat an LLM as one more high level language. (And where has engineering gone? Engineering is the difference in the thinking itself, in knowing what to make and in connecting technologies and developing them.)

this name stays in this browser. it is not sent anywhere.

I love vibecoding

Tony Stark is a vibecoder with unlimited tokens. As far as I see, internships and projects are presented as though they belong to certain people. What gets inherited is not only money. You can call it inherited vision, you can call it social climbing, and there is even a niche for it now, something like a class based networking hub founder. What will happen soon is that somebody starts a business with a Claude Pro account and has graduates of good schools working underneath them, when those same students could have started that company with their pocket money. Nevertheless, if it is not rocket science and it

is not tank design, I do not think any work that needs no large capital is as big and as irreplaceable as it is claimed to be. Your application was rejected, so say that you could do this too and carry on your way.

Everyone should benefit from technology, and not only for business and money, but for making products for your own needs. As I believe from Plato, no one does wrong willingly, and nobody writes bad software on purpose either. Outside academic purposes, CS is no different from being able to sew a button on. Everyone should learn it in order to make the ideas they dream of real, and most importantly, everyone can, because I believe that for once in history anyone can build anything.

I have models write code for me, and I will have them write code while I am writing this book. Some things inside my own projects are plainly wrappers and I am not going to hide that from you while asking you to be honest about yours.

What matters is whether anything stands behind the code. Think about the last time your terminal told you every test passed. It told you that some tests ran and none of them complained, and nothing about

what should I call you?

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Who am I?

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Actually I was a coder before CS took over, starting with HTML at 11 and a lot of Stack Overflow. Now I vibecode anyway. (Okay, what I really do is LLM systems engineering, which means I connect systems and engineer them.)

ir-globe draws what 198 countries do to each other and it has been going for months without me touching it, on infrastructure that costs 0 dollars a month. stitchu takes a photograph of a garment and gives back a sewing pattern, and it has a gate that decides whether that pattern can actually be sewn, which I wrote after printing patterns that passed every test I had and could not be sewn at all. lulumelon measures what language models say about a brand and prints a range rather than a single number, because on its first live run the honest interval ran from 0 to 33,3. rabadon is a gate system for agents, written in C++ with no dependencies, and it exists because an agent once told me it had finished a job it had not finished.

Each of those started as something I got wrong, then found, then fixed, and the fix is what this book hands over.

How should you read this book?

The order is semantic, and I spent a full week reading and thinking before I settled it. I started from localhost, because that is the joke I kept seeing. Writing it made me realise the link only fails if you believe a page is a place, so that subject had to come first. Writing that one made me realise I was talking about files without ever saying what a folder is, so the terminal came next.

Localhost then required `npm run dev`, because once you know whose machine answers you want to know what is running on it. That one required `npm run build`, because the thing you would put on another machine is not the thing running on yours. After that came the backend, because a program on another machine is what the whole book has been walking towards. The backend required data, because a backend keeps something. Data required `.env` files and API keys, because the moment something is kept, whatever reaches it matters.

Those chapters are grouped into five parts. 101 is what you are actually holding. 201 gets your project off your own computer. 301 is where people touch, system design. Even the thing you can stretch the screen in any to

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By the way, new tabs keep opening in my head, and a chapter about one thing wakes up an idea about something else entirely. Those ideas do not belong inside the chapter, so they go in the appendix, and terms and small technical things that do not fit there go in the footnotes. Both sit at the bottom of the page, so anything marked with a star has a note waiting for it.

The appendix is also where the writing about product management and the startup world will go. Companies still run events to hand out the kind of knowledge you can only get through experience, and then they turn creative people away with reasons on the level of you are a Gemini, that is why you were eliminated. To tell you the truth, I do not think anybody is as well intentioned as I am, and people hold a strange principle about not sharing good things. You can steal my ideas. There is sand in the sea and there are ideas in me, so passing on what I know is nothing but a happiness for me.

Thanks

Years ago the yazbel documentation made me interested in computer science, and I changed my department. Even though we have not been writing code by hand for the last year, it gave me a love of CS that no undergraduate degree could have given me. We used to shorten loops with algorithm analysis, and now we do the same thing inside workflows. The language changed and the tool changed, while the purpose and the logic did not, so I still think this motion in CS is a matter of perspective.

So when I am no longer a sweet student but a businesswoman, I hope this book will have enlightened somebody, entertained somebody, or made somebody think.

Two heads are better than one, so whatever work you are in, try your own idea too. My endless respect to everyone who is dreaming and working for something.

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what is the thing on your screen?

A page is not a place. It is a delivery that happens again every time somebody looks.

The first thing you do with a coding agent is ask it for a website, and a minute later there is one. The terminal prints an address. You open it and the thing you described is sitting in the browser looking finished.

Ask where that thing lives and the answer comes back as in the browser or on my computer or on the internet. All three feel fine right up until something breaks, and then none of them tell you where to look.¹

1

A book note. A model takes the easiest road. It gave you localhost with no backend because that is the shortest way to something that runs.

what should I call you?

What arrives

this name stays in this browser. it is not sent anywhere.

Make a file named index.html with one line inside.

```
<h1>hello</h1>
```

Open a terminal in that folder and run `python3 -m http.server 8000`. Go to `localhost:8000` in your browser. The word hello is there in large letters, arriving from a server small enough that nothing can hide inside it.

Now right click on the page and choose view source. What comes up is the line you typed, character for character.

Change the `h1` to a `p` and refresh. The same word comes back small. Nobody sent you a smaller word. You changed one instruction and the browser drew the letters differently, because what travelled to it was never a page. It was a set of instructions for a program that carries them out.

How does the browser learn about the second file?

One file is not how anything real arrives. Next to index.html make a file called style.css containing `h1 { color: crimson; }` and mention it from the page.

```
<link rel="stylesheet" href="style.css">
<h1>hello</h1>
```

Refresh and the word is red.

Your browser had no way of knowing that style.css existed. The only file it had been given was index.html. It read that file and came to a line mentioning a stylesheet. Only then did it go back to the server for a second file. The page tells the browser what else to fetch, one discovery at a time.

Open the developer tools with Cmd and Option and I on a Mac or F12 on Windows. Click the Network tab and refresh with the panel open. Two rows come up, index.html first, and style.css second, and they ca

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Where does the page go when the server stops?

'The page is on my computer, so it will still be there,' you may be thinking. In the Network panel there is a checkbox marked Disable cache. Tick it so the browser stops keeping copies of its own. Now go to the terminal and stop the server with Ctrl and C, then refresh the page.

The page is gone and there is no error about your HTML, because your HTML never arrived. Nothing had been sitting in the browser waiting for you.

Start the server again and refresh. Hello comes back in red exactly as before, drawn from nothing. A page is a delivery that happens again every time somebody looks.

What does this look like in a real project?

Open the project you have been working on and refresh it with the panel open. The same list appears, only far longer. The document sits at the top. Everything the document mentioned sits under it, and under those sits everything they mentioned in turn.

At the bottom of the panel there is a total size. Every visitor downloads that much before they see anything at all, out of their connection and their phone plan.

That leaves one question for the rest of the book. Some program on some machine sent those files. On the hello page you know which one, because you started it yourself and it is still sitting in your terminal. On your real project you have probably never thought about it.

CHECK

Open the Net panel on the page you just created and refresh it.

How many files

How many k

Which file in the list is the one you just created? It is always one. Usually a font you stopped using months ago or a library that arrived attached to something else. Nothing announced it, because it was mentioned by a file that was itself mentioned by a file.

Nothing needs removing today. It has been going out to every visitor since the day it appeared.

You just started a server inside a folder without my having said what a folder is, and that gap has to close before we can ask whose machine your files come from. So the terminal and the folder come next.

why can your deploy not find your files?

The same command run one level up serves nothing. Most deploy failures start here.

You ran a command inside a folder a moment ago and a server appeared, and I never said which folder it was reading or why that mattered. The same command run from one step higher up would have served nothing.

A whole family of failures comes out of that. The build that works on your machine and returns file not found on the deploy is one, the image that loads locally and 404s in production is another. In every one of them a program is standing somewhere you did not expect, reading a path that means something different from there.

Where are you

what should I call you?

Open a terminal window, use the one in the top left black box.

A command is standing somewhere and where you are standing changes what it does.

```
pwd
```

Print working directory. It answers with the absolute path of where you are standing right now.

```
ls -F
```

Plain ls lists files and folders together with no way to tell them apart, and -F puts a slash on the end of every folder name. Do not try to work it out from the name. README has no dot and is a file, while a folder called backup.old has one and is still a folder.

```
cd Desktop
```

```
cd ..
```

That is the whole of navigation. Three commands and one convention, and they work the same way on every Mac and every Linux machine and inside the terminal your Windows tools installed.

this name stays in this browser. it is not sent anywhere.

```
/Users/damla/Desktop/project
```

There are two kinds.

Why does this decide whether things work?

Every build step, every deploy tool and every error message you will read for the rest of your life assumes you know where you are standing.

File does not exist almost always means a relative path was read from a folder you were not expecting. The file was there all along, and the program was standing somewhere else when it went looking for it. You saw exactly that a minute ago with a server one level too high.

Something works on your machine and dies on the deploy for the same reason. You typed the command while standing inside the project folder. The other machine runs it while standing wherever it happens to start, so the relative path points at a place that does not exist. Chapter 16 is where this bites hardest.

MIT teaches a free course for exactly this gap called The Missing Semester of Your CS Education. Julia Evans draws comics about the shell that are the friendliest thing on the subject.

what should I call you?

CHECK

Open a terminal

Get to your project

comes back and

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When you arrive, which folder in that list can you not describe the contents of? There will be one, usually node_modules or dist or .next.

Go into it and look, changing nothing. The next time something breaks that folder is going to appear in the error message (which should not be the first time you see the name).

Next comes localhost, and the reason a link that opens instantly for you does not open for your friend.

why does your link only open for you?

Your machine answers its own name. Your friend's machine answers its own, and finds nothing.

This chapter is about famous localhost:3000. Jokes about it have been going around the internet for a long time. Many people believe CS is dead because they built a website hosting on local. You click your own link and it opens. Your friend clicks it and it does not. Why does it happen, what is localhost:3000, and how to fix this issue?

What does localhost mean?

It begins with localhost:3000 or 127.0.0.1:3000, which are the same address with different names. The number after the colon is the port number. The address is the building, the port is the apartment inside it. One machine runs many programs and each waits at its own number. 3000 is a convention development tools picked, nothing more. The port is fine. The building is the problem, because you gave your friend the name every building uses for itself.

what should I call you?

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Can your own phone open it?

Open the terminal and cd into any folder that has a file or two in it. Run this.

```
python3 -m http.server 8000 --bind 127.0.0.1
```

This starts a small web server. The flag tells it to answer only the machine it is running on. Open localhost:8000 in the browser and you see the folder's files listed. Now type the same address into your phone's browser. Nothing comes back, even though the phone is on the same wifi. The phone went looking for a program at port 8000 on itself.

Now stop the server with Ctrl+C and start it again without the flag.

```
python3 -m http.server 8000
```

The flag was doing all the work. 127.0.0.1 told the server to listen on the loopback address alone, so requests arriving over wifi were never answered. Without the flag this tool listens on 0.0.0.0, which is not an address you connect to. It means every network interface the machine has, including the wifi card. A request from the phone is now accepted the same as one from the machine itself. You will see 0.0.0.0 again in Docker logs and in cloud deploy output and it always means the same thing.

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How far does the local network reach?

'So I drop the flag and I am live,' you may be thinking. What you have is a server reachable from the same wifi. 192.168.1.24 is an address inside your home network and means nothing outside it, so a friend in a cafe or a customer on mobile data cannot reach it. And when the laptop lid closes, nothing is running anywhere.

A product has to answer while you are asleep, so the program has to run on a machine that stays on and has an address the internet can route to. Putting it there is called deploying, and that is chapter 16. ir-globe answers at three in the morning because it is not on my laptop.

CHECK

Look at the address bar of the project you are proudest of. If it begins with localhost or 127.0.0.1, the number of people who can open it is 1. Then close the laptop. Turn off wifi on your phone so you are on mobile data and try to open the project. Whatever comes back is the answer to whether it exists yet.

After that come the two commands you run without reading them. Only one of `npm run dev` and `npm run build` makes a thing you can deploy.

what should I call you?

this name stays in this browser. it is not sent anywhere.

what is the difference between npm run dev and npm run build?

One builds pages in memory for you. The other writes the folder your users get.

Localhost turned out to be a name your machine uses for itself. The other half of the address is still unexplained, because something is answering at that door, and in a framework project that something is npm run dev.

You have run that command a thousand times without wondering what it is. Sitting next to it in the same file there is npm run build. It takes much longer and produces a folder you have probably never opened. That folder is the part your users get.

What is run

what should I call you?

What to be asked to be chapter 1. This sent anywhere.

Go and C. Refresh the browser. The site is gone and the error is a refused connection, which means nothing is listening at that door any more.

Now start it again and open the project folder, looking for the HTML that arrived in the browser. There is no such file. Your project has a src folder full of components and not one of them is a page. Nothing on the disk matches what you saw in view source.

The dev server built that page in memory the moment your browser asked for it and then let it go. Its whole job is to read the source, work out what the browser needs right now and hand it over. Then it watches the files so it can do the same thing the instant you save. Nothing was written down, so a save appears in the browser in under a second.

What does the build do?

```
npm run build
```


It takes a while, where the dev server started instantly. When it finishes there is a new folder called dist on Vite and .next on Next.js. Go in there with the commands from chapter 2.

```
cd dist
ls -F
```

The HTML that did not exist a minute ago is in there. So are the scripts, with names like index-4f3a9c.js. Those characters in the middle are there so a browser can safely keep an old copy without mistaking it for the new one after you deploy. The CSS from every component in the project has been gathered into one file.

Your source is written in whatever shape lets you work. This folder is written for browsers, in one shape and once. The dev server had been standing in for it on demand without ever writing a word of it to disk.

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```
cd dist
python3 -m http.server 8000
```

Open localhost:8000. That is your site, coming out of a plain file server that has never heard of your framework. It is the same server that handed you one line of HTML in chapter 1.

Why do the two behave differently?

'Same code, same project, so it will behave the same way,' you may be thinking. Open one of those scripts in dist and read it. Your function names are gone, replaced by single letters. The spaces are gone. The whole file is one line.

The dev server is optimised for you. It starts instantly and refreshes instantly, keeps your variable names so errors stay readable, and squashes nothing. The build is optimised for the visitor, so it makes the smallest files it can, squashes everything together, drops code that nothing uses and shortens every name it can reach.

Squashing changes behaviour. Code that quietly relied on a function still having its own name breaks, because the name is gone and you just read the file where it went missing. A value read while the build is running gets frozen into the file, so changing it on the server afterwards changes nothing until you build again, and chapter 13 is about the trouble that causes. A file the dev server found by walking your folder is missing from the output entirely if nothing imported it properly.

So a project can run beautifully for weeks and fall over on the first deploy without a line of source having changed. When somebody says it works on my machine they are usually telling the truth, because their machine was running the friendly version.

Before you deploy, run the build and serve the output locally the way you would serve it in production. If you are sitting at a computer that for some reason is broken in a way that you can't see, you are sitting at a computer that for some reason is broken in a way that you can't see.

what should I call you?

this name stays in this browser. it is not sent anywhere.

CHECK

Run `npm run build` on your own project and let it finish.

How long did it take? Whatever that number is, it happens again on every deploy.

What is inside the folder? Go in with `cd dist` or `cd .next` and run `ls -F`, then find the HTML your source never contained.

Then serve it with `python3 -m http.server 8000` and use your own site coming out of a file server that knows nothing about your framework. If something works under `npm run dev` and not in there, you have found a real bug months before a stranger would have found it for you.

Chapter 5 goes to the shop front and the till. What a backend is, and why some things can never live in the files you just looked at however well you hide them.

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sent anywhere.

why does the till stand in a different room?

Everything delivered to the browser belongs to whoever is standing there.

Everything so far has happened in one room. Files arrive and the browser draws them. You have now read your own project the way a visitor reads it. A shop has a second room behind the counter where the till and the keys are, and nobody has to be told why. In a project both rooms look like folders sitting next to each other in the same editor, so the till ends up out on the shop floor.

Which room is the browser?

The browser is standing in the room of whoever is standing there. You can click a button. That will not happen. That is linking. Open a button marked with a script at the bottom of the page. The one line becomes readable code again. Squashing a script makes the file smaller and it hides nothing.

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What is a backend?

A backend is a program on another machine that answers requests. It runs somewhere else and listens at a door. You talk to it by sending exactly the kind of request you watched arrive in chapter 1.

The customer orders and the kitchen cooks. The customer does not come into the kitchen, because the kitchen is a different room. A script running in the browser is a locked cupboard standing in the dining room, and the customer has all evening.

What is an API?

An API is the menu. A menu tells you what you may order and in what form. An API tells you which requests the kitchen accepts, meaning this address with this method and these fields. Anything not on the menu is refused. Writing one is the act of deciding what strangers may ask you for.

The verbs on that menu are the methods. GET fetches something, POST sends something new, PUT replaces, DELETE removes. A GET is understood everywhere to only fetch and search crawlers act on that understanding. That is how a link which deletes a row gets triggered by a crawler that was only looking around.

What do the numbers mean?

Every answer comes back with a status number and MDN carries the definition.

200 means you perfectly cooked the food. A 401 means it is not known where the recipe is, so you need to cook it.

The status numbers are separate systems with separate rules, and joining them apart is the whole of chapter 26.

what should I call you?

this name stays in this browser. it is not sent anywhere.

Can you break your own rule?

Find something in your own project that the page refuses to let you do. A submit button that stays disabled until the form is valid, a field with a maximum length, a section that only appears when you are logged in. Anything the page decides.

Open the developer tools and go to the Console tab. Then type one of these, matching whichever rule you found.

```
document.querySelector("button[disabled]").disabled = false
document.querySelector("input[maxlength]").maxLength = 9999
```

The button is live. The field takes as much as you want. You did not find a hole in your code, you retyped a line of it, and every visitor has the same console you just used.

That is why a check written into your page is a suggestion. Your script runs on their machine. A check that holds has to run in a room they cannot reach, which is the room this chapter is about.

Where does the key go?

Into the kitchen and never into a request the customer can read.

stitchu sends a photograph to a vision model and that model charges per call. The key that pays for it belongs in the till, so the browser never holds it. The photograph goes to a Cloudflare Worker, the Worker holds the key and calls the model, and the answer comes back. Anyone can open the Network panel on stitchu and read every request that leaves their machine and find nothing in them to take.

what should I call you?

what should I call you?

```
this name stays in this browser. it is not
sent anywhere.
```

CHECK

Open your own project with the Network panel open and use it until requests start appearing.

Where is each request going? Read the address of each one. If it goes straight to somebody else's service then your browser is talking to that service directly and everything in that request is public.

What in your code costs money or grants access? An API key, a database URL, an admin token. Where does it live right now?

If it reached the browser then it is public, and the next two chapters are about moving it.

Chapter 6 goes to where data lives, and why a file works perfectly until two people use your project at the same time.

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